

Diagnostic Test — Calculus I (MATH1013)

2026-07-05

If you get stuck, write down which step broke. That note is the most valuable output.

Part 1 — Limits & Continuity

- 1.1 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.
- 1.1 Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$.
- 1.3 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 1}{2x^2 + 7}$.
- 1.4 Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$.
- 1.2 Let $f(x) = x^2$ for $x \leq 1$ and $f(x) = ax + 1$ for $x > 1$. Find a so that f is continuous at $x = 1$, and state which theorem guarantees a root of $g(x) = x^2 - x - 1$ in $[1, 2]$.

Part 2 — Derivatives

- 2.1 Use the limit definition (not rules) to find $f'(x)$ for $f(x) = \frac{1}{x}$.
- 2.2 Differentiate $f(x) = (3x^2 + 1)(x^3 - 2x)$.
- 2.3 Differentiate $f(x) = (2x + 1)^5$.

Part 4 — Integral Calculus

- 4.1 Find $\int (4x^3 - 6x + 5) dx$.
- 4.4 Find $\int 2x(x^2 + 1)^4 dx$.